

Materials Science Recommendation Report

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User Request:

Design a low-cost porous composite for atmospheric water harvesting using activated carbon, porous silica, hygroscopic salt, stabilizers, and polymer binder

Material Category:

Atmospheric Water Harvesting Material

Target Application:

Research and development guidance for AWH material design

Composition:

Component	Ratio	Percent
Activated carbon or porous carbon	0.300	30.0%
Porous silica or silica gel	0.250	25.0%
Aluminum oxide, inorganic stabilizer, or clay	0.150	15.0%
Calcium chloride or controlled hygroscopic salt	0.150	15.0%
Cellulose, polymer binder, or structural polymer	0.100	10.0%
Titanium dioxide, biochar, or carbon black (photothermal additive)	0.050	5.0%

Recommended Processing / Fabrication Method:

- 1. Powder Mixing and Homogenization:
- 2. Binder Preparation:
- 3. Paste Formation:
- 4. Shaping and Molding:
- 5. Drying:
- 6. Activation Cycling (optional):
- 7. Evidence Boundary:

Category-Specific Parameters & Targets:

Relative Humidity Range:	40-90% RH (optimal moisture capture window)
Water Uptake Target:	0.3-0.8 g water per g dry material
Adsorption Time:	4-12 hours under ambient conditions
Desorption Temperature:	50-80 degC (thermal or solar regeneration)
Regeneration Method:	Sunlight-assisted heating or low-grade thermal (waste heat compatible)
Cycling Target:	50-100 wet/dry cycles minimum (durability benchmark)
Salt Leaching Test Method:	Immersion test + chloride/conductivity measurement per cycle
Collected Water Quality Test:	pH, TDS, conductivity, trace metals (ICP), microbial screening

Validation Plan:

Water Uptake Capacity:	Quantified at 40%, 60%, 80% RH via gravimetric analysis
Rh Working Range:	Minimum RH (material saturation limit) and maximum RH (hygroscopic plateau)
Adsorption Kinetics:	Time to reach 90% equilibrium saturation (breakthrough curve)
Desorption Kinetics:	Time to release >80% absorbed water during thermal cycle
Regeneration Energy:	Thermal input (kWh) required per kg water recovered
Cycling Durability:	Retained water uptake (%) after 50 and 100 wet/dry cycles
Salt Leaching:	Salt mass balance per cycle; acceptable levels per end-use water standard
Collected Water Quality:	pH, TDS, conductivity, chloride (mg/L), trace metals, and microbial count per appli

Safety & Regulatory Tests:

- 1. Salt leaching analysis
- 2. Collected water quality (pH, TDS, conductivity, microbial)
- 3. Trace metal analysis (ICP)
- 4. Cycling durability

Scientific Dataset Verification Summary:

External dataset verification was not completed. Report relies on internal category registry and preset library.

Components Verified: 0/0

Materials/Properties Found: 0

Supporting Literature Found: 0 papers

Evidence: External dataset verification was not completed.

Verification Summary:

Category Selected: Atmospheric Water Harvesting Material

Stage 1 - Category Keyword Verification: PASS (95% match)

Stage 2 - Preset Field Compatibility: PASS

Stage 3 - Scientific Consistency: PASS

Export Status: Passed

DISCLAIMER: All material parameters, compositions, and performance targets in this report are AI-generated planning defaults based on materials science knowledge. These parameters DO NOT demonstrate proven water-harvesting performance, potable-water safety, commercial readiness, or

suitability for any specific application. All recommendations are CONDITIONAL upon rigorous laboratory validation, field testing, compliance with applicable water-quality standards, and consultation with qualified materials engineers and water-quality specialists. This report is for research and development guidance only.